Astrophysical Fluid Dynamics

Lecture notes for ASTRO 442

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Contents

Preface . . . 5
How to use these notes . . . 5
Accessibility . . . 5
License . . . 5

Schedule . . . 7

I Foundations

1 Fluids as Continua . . . 11

2 Thermodynamics Refresher . . . 13

3 Core-Collapse Supernovae . . . 15

4 The Euler Equation . . . 17

5 Eulerian and Lagrangian Descriptions . . . 19

6 Conservation Laws . . . 21

7 The Gravitational Potential . . . 23

8 Hydrostatic Equilibrium and the Virial Theorem . . . 25

II Waves, Shocks, and Outflows

9 Sound Waves and the Doppler Effect . . . 29

10 Waves in Stratified Atmospheres . . . 31

11 Shock Waves . . . 33

12 Radiative Shocks . . . 35

13 Blast Waves . . . 37

14 Stellar Winds and Spherical Accretion . . . 39

III Instabilities, Viscosity, Disks, and Turbulence

15 Fluid Instabilities I: Convection 43
16 Fluid Instabilities II: Rayleigh–Taylor and Kelvin–Helmholtz . 45
17 Fluid Instabilities III: Thermal and Gravitational Instabilities . 47
18 Diffusion and Viscous Flows 49
19 Accretion Disks I: Structure and Viscous Evolution 51
20 Accretion Disks II: Steady Disks and Spectra 53
21 Turbulence 55

IV Plasma and Magnetohydrodynamics

22 Introduction to Plasma 59
23 The MHD Equations 61
24 Flux Freezing 63
25 Magnetic Tension and Hydromagnetic Equilibria 65
26 MHD Waves 67
27 The Magnetorotational Instability 69

Appendices 71
A Vector Calculus 73
Bibliography 75

Preface

These are the lecture notes for ASTRO 442, an introduction to the fluid dynamics and magnetohydrodynamics of astrophysical systems. They are written to be self-contained: the textbook, Clarke and Carswell's *Principles of Astrophysical Fluid Dynamics* [1], is optional further reading. Each chapter ends with pointers to the relevant sections of the textbook.

> **i** Note
>
> The notes are being prepared for the Spring 2027 semester. Chapters marked "in preparation" will be completed before the corresponding lecture; see the Schedule.

How to use these notes

- **One chapter per lecture topic.** The Schedule lists which chapter goes with each lecture.
- **Everything is numbered by chapter.** Equation (9.14) is the fourteenth numbered equation of Chapter 9, and Exercise 8.4 is the fourth exercise of Chapter 8. When you ask a question about the notes, quote these numbers.
- **In-class worksheets** appear near the end of each chapter, followed by the **exercises**. Homework assignments on Canvas list the exercises to solve, for example "Exercises 2.1, 7.3".
- **Optional material** is collapsed under "Going further" boxes.

Accessibility

These notes are designed to meet the Web Content Accessibility Guidelines (WCAG) 2.1, level AA. Mathematics is rendered with MathJax, which supports screen readers, speech, and braille, and lets you explore an equation term by term: click on an equation or focus it with the keyboard, then use the arrow keys. Every figure has a text description.

If something in these notes is not accessible to you, please let me know.

License

The text and figures are licensed under CC BY-NC-SA 4.0. Code is licensed under the MIT license. The source is on GitHub.

Schedule

Lecture schedule for Spring 2027 and the chapter that goes with each lecture. Lectures are on Tuesdays and Thursdays. Homework assignments and due dates are posted on Canvas.

Lecture	Date	Topic	Notes
1	Tue Jan 12	Fluids as continua, continuity equation	Chapter 1.
2	Thu Jan 14	Energy equation, thermodynamics	Chapter 2.
3	Tue Jan 19	Core-collapse supernovae	Chapter 3.
4	Thu Jan 21	Vector calculus	Appendix A.
5	Tue Jan 26	Euler equation; Eulerian and Lagrangian descriptions	Chapter 4., Chapter 5.
6	Thu Jan 28	Conservation laws	Chapter 6.
7	Tue Feb 2	Gravitational potential	Chapter 7.
8	Thu Feb 4	Hydrostatic equilibrium, virial theorem, star formation	Chapter 8.
9	Tue Feb 9	Sound waves, acoustic power	Chapter 9.
10	Thu Feb 11	Doppler effect	Chapter 9.
11	Tue Feb 16	Sound waves in stratified media	Chapter 10.
12	Thu Feb 18	Shocks and shock adiabatics	Chapter 11.
13	Tue Feb 23	Radiative shocks	Chapter 12.
14	Thu Feb 25	Blast waves	Chapter 13.
15	Tue Mar 2	**Midterm I**	
16	Thu Mar 4	Stellar winds and spherical accretion	Chapter 14.
	Mar 9–11	*Spring break*	
17	Tue Mar 16	Fluid instabilities I	Chapter 15.
18	Thu Mar 18	Fluid instabilities II	Chapter 16.
19	Tue Mar 23	Introduction to plasma	Chapter 22.
20	Thu Mar 25	**Midterm II**	
21	Tue Mar 30	Fluid instabilities III	Chapter 17.

Lecture	Date	Topic	Notes
22	Thu Apr 1	Diffusion and viscous flows	Chapter 18.
23	Tue Apr 6	Accretion disks I	Chapter 19.
24	Thu Apr 8	Accretion disks II	Chapter 20.
25	Tue Apr 13	Turbulence	Chapter 21.
26	Thu Apr 15	MHD equations	Chapter 23.
27	Tue Apr 20	Flux freezing	Chapter 24.
28	Thu Apr 22	Magnetic tension	Chapter 25.
29	Tue Apr 27	MHD waves	Chapter 26.
30	Thu Apr 29	Magnetorotational instability	Chapter 27.

I Foundations

1 Fluids as Continua 11

2 Thermodynamics Refresher 13

3 Core-Collapse Supernovae 15

4 The Euler Equation 17

5 Eulerian and Lagrangian Descriptions 19

6 Conservation Laws 21

7 The Gravitational Potential 23

8 Hydrostatic Equilibrium and the Virial Theorem 25

1. Fluids as Continua

> **Note**
>
> This chapter is being ported from the 2025 lecture notes.

2. Thermodynamics Refresher

> **Note**
>
> This chapter is being ported from the 2025 lecture notes.

3. Core-Collapse Supernovae

> **Note**
>
> This chapter is being ported from the 2025 lecture notes.

4. The Euler Equation

> **Note**
>
> This chapter is being ported from the 2025 lecture notes.

5. Eulerian and Lagrangian Descriptions

> **Note**
>
> This chapter is being ported from the 2025 lecture notes.

6. Conservation Laws

> **Note**
>
> This chapter is being ported from the 2025 lecture notes.

7. The Gravitational Potential

> **Note**
>
> This chapter is in preparation.

8. Hydrostatic Equilibrium and the Virial Theorem

> **Note**
>
> This chapter is being ported from the 2025 lecture notes and expanded.

II

Waves, Shocks, and Outflows

9 Sound Waves and the Doppler Effect . 29

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13 Blast Waves 37

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9. Sound Waves and the Doppler Effect

> **Note**
>
> This chapter is being ported from the 2025 lecture notes and expanded.

10. Waves in Stratified Atmospheres

> **Note**
>
> This chapter is being ported from the 2025 lecture notes and expanded.

11. Shock Waves

> **Note**
>
> This chapter is in preparation.

12. Radiative Shocks

> **Note**
>
> This chapter is in preparation.

13. Blast Waves

> **Note**
>
> This chapter is in preparation.

14. Stellar Winds and Spherical Accretion

> **Note**
>
> This chapter is in preparation.

III Instabilities, Viscosity, Disks, and Turbulence

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18 Diffusion and Viscous Flows 49

19 Accretion Disks I: Structure and Viscous Evolution 51

20 Accretion Disks II: Steady Disks and Spectra 53

21 Turbulence 55

15. Fluid Instabilities I: Convection

> **Note**
>
> This chapter is in preparation.

16. Fluid Instabilities II: Rayleigh–Taylor and Kelvin–Helmholtz

> **Note**
>
> This chapter is in preparation.

17. Fluid Instabilities III: Thermal and Gravitational Instabilities

> **Note**
>
> This chapter is in preparation.

18. Diffusion and Viscous Flows

> **Note**
>
> This chapter is in preparation.

19. Accretion Disks I: Structure and Viscous Evolution

> **Note**
>
> This chapter is in preparation.

20. Accretion Disks II: Steady Disks and Spectra

> **Note**
>
> This chapter is in preparation.

21. Turbulence

> **Note**
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IV Plasma and Magnetohydrodynamics

22 Introduction to Plasma 59

23 The MHD Equations 61

24 Flux Freezing 63

25 Magnetic Tension and Hydromagnetic Equilibria 65

26 MHD Waves 67

27 The Magnetorotational Instability . . 69

22. Introduction to Plasma

> **Note**
>
> This chapter is being ported from the 2025 lecture notes.

23. The MHD Equations

> **Note**
>
> This chapter is being ported from the 2025 lecture notes.

24. Flux Freezing

> **Note**
>
> This chapter is being ported from the 2025 lecture notes.

25. Magnetic Tension and Hydromagnetic Equilibria

> **Note**
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> This chapter is being ported from the 2025 lecture notes.

26. MHD Waves

> **Note**
>
> This chapter is being ported from the 2025 lecture notes.

27. The Magnetorotational Instability

> **Note**
>
> This chapter is being ported from the 2025 lecture notes.

Appendices

A. Vector Calculus

> **Note**
>
> This chapter is in preparation.

Bibliography

[1] C. Clarke and B. Carswell, *Principles of Astrophysical Fluid Dynamics*. Cambridge: Cambridge University Press, 2007.